

OCI Gen2 106G NRZ — SerDes TX Discussion Outline

For: Cadence (CDNS) L250 SerDes Team
From: Lightmatter
Purpose: Alignment on TX FIR requirements and interface spec ahead of slide development
Status: Preliminary — for discussion

1. Context

Lightmatter is designing an OCI Gen2 coherent photonic interconnect running **106.25 GBaud NRZ** per lane over 500 m SMF. The TX SerDes (CDNS L250) drives an integrated micro-ring modulator (MRM). This is an optical modulation application — the channel and equalization requirements differ significantly from a standard electrical SerDes application.

Key differences from a standard backplane/copper SerDes deployment:

- Driver sees a **lumped RC load** (MRM electrode: $C \approx 10\text{--}30\text{ fF}$, $R \approx 10\text{--}30\text{ }\Omega$) — no transmission line, no $50\text{ }\Omega$ termination
- Channel ISI is dominated by the **MRM Lorentzian optical response** (BW target $\geq 60\text{ GHz}$), not a PCB trace
- **No RX-side FFE or DFE** — the TX FIR must pre-compensate the full channel (MRM + RX TIA) with 3 taps only
- **Asymmetric rise/fall correction** is required to cancel MRM optical nonlinearity

2. TX FIR Requirements

2.1 Tap Configuration

- **3 taps:** pre-cursor (c_{-1}), main cursor (c_0), post-cursor (c_{+1})
- Normalization: $|c_{-1}| + c_0 + |c_{+1}| = 1$
- **Working design point:** $c_{-1} = -0.10$, $c_0 = 0.65$, $c_{+1} = -0.25$

The post-cursor tap is dominant and compensates the MRM Lorentzian post-ISI **and** the RX TIA response. Because there is no RX FFE, the TX FIR carries all equalization responsibility.

Discussion topic for CDNS: Can L250 support coefficient optimization targeting a photonic (MRM + optical TIA) channel model rather than a standard electrical channel? What is the expected convergence path?

2.2 Coefficient Range and Resolution

Tap	Range	Step size	Notes
c_{-1} (pre)	0 to $-0.30\times$	3-bit (0.043/step)	Group delay from TIA
c_0 (main)	derived from normalization	—	
c_{+1} (post)	0 to $-0.30\times$	3-bit (0.043/step)	MRM + TIA ISI; needs -0.25

Discussion topic: Is -0.25 reachable with 3-bit resolution (step 0.043)? Nearest 3-bit value: -0.258 (6 steps). Confirm quantization error is acceptable.

2.3 Asymmetric Rise/Fall Correction

MRM transfer function $T(V)$ is nonlinear (Lorentzian) — the optical edge into resonance is faster than the edge out of resonance. This creates duty-cycle distortion (DCD) in the optical eye. The driver must apply **independent, asymmetric slew rate control** on rise vs fall.

Discussion topic for CDNS: Does L250 support independent tr/tf adjustment (not just symmetric peaking)? What is the control granularity? Can this be software-programmable per-channel at runtime (temperature-tracking requirement)?

3. Electrical Interface Spec (Driver Output)

Parameter	Target	Notes
Differential output swing	0.5–1.5 Vpp diff	1.0 Vpp typical for ER $\approx$ 4.5 dB
Output impedance / return loss	N/A	No transmission line — lumped RC load only
Rise/fall time (20-80%)	≤ 6 ps	At MRM electrode; after TX FIR
Baud rate	106.25 GBaud	OCI Gen2

4. Jitter Budget ($\text{BER} = 1 \times 10^{-12}$)

Target total jitter at TP2 (optical fiber attach):

Component	Budget
DJ_pp total	≤ 2.00 ps_pp (21% UI)
σ_{RJ}	≤ 0.15 ps_rms
TJ at $\text{BER} = 1\text{e-}12$	≤ 4.11 ps_pp (44% UI)

Discussion topic for CDNS: What is L250's TX output jitter at 106.25 GBaud with 3-tap FIR active? Is the $\sigma_{\text{RJ}} \leq 0.15$ ps spec achievable from the PLL at this baud rate?

5. Open Questions for CDNS

- FIR coefficient adaptation:** Does L250 support an adaptive/LMS mode for TX FIR, or coefficient loading only? For this application, startup adaptation to an MRM+TIA channel model is needed.
- Asymmetric peaking control:** Independent programmable tr/tf slew — available? Granularity?
- Rise time spec margin:** What is L250's nominal electrical rise time at 106 GBaud with 3-tap FIR? Do we have margin to the 6 ps target?
- Jitter characterisation:** Can CDNS provide σ_{RJ} and DJ_pp data at 106.25 GBaud for L250?

5. **Supply sensitivity (PSIJ):** What is expected PSIJ contribution from SerDes supply coupling at expected supply noise frequencies? Target ≤ 0.30 ps_pp PSIJ.

6. **SS corner performance:** L250 TX output BW at SS process corner — does rise time stay ≤ 6 ps?

6. Suggested CDNS Slide Topics

- L250 TX FIR: tap count options, coefficient range/resolution at 106 GBaud
 - Asymmetric tr/tf capability and software programmability
 - TX output jitter characterisation (DJ, σ_{RJ}) at 106 GBaud
 - TX output swing range and programmability
 - Channel adaptation flow for non-standard (optical) channel
 - PSIJ / supply coupling characterisation
 - SS corner electrical performance data
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7. Next Steps

Action	Owner	Timing
CDNS to confirm L250 3-tap coefficient range at 106G	CDNS	Before next sync
CDNS to share TX jitter (DJ, RJ) characterisation data at 106 GBaud	CDNS	Before next sync
CDNS to confirm asymmetric tr/tf adjustment capability	CDNS	Before next sync
Lightmatter to share MRM + TIA channel model for FIR optimisation	Lightmatter	After MRM S21 measurement
Joint FIR coefficient optimisation session	Both	TBD